

Gorkha Northern VDCs Study & Mud Mortar Additive Research update

Presented by : CRS

Introduction

CRS: Gorkha Recovery and Resilience Program (Oct 2016 - Sep 2019)

Program focus on (Shelter, WASH & Livelihood)

Working Area : 13 VDCs and Gorkha Municipality

Northern VDCs : 4

This presentation is based on the field visit & study conducted by CRS to provide input to its Shelter Nepal Program by observing and assessing the remote situation in the North.

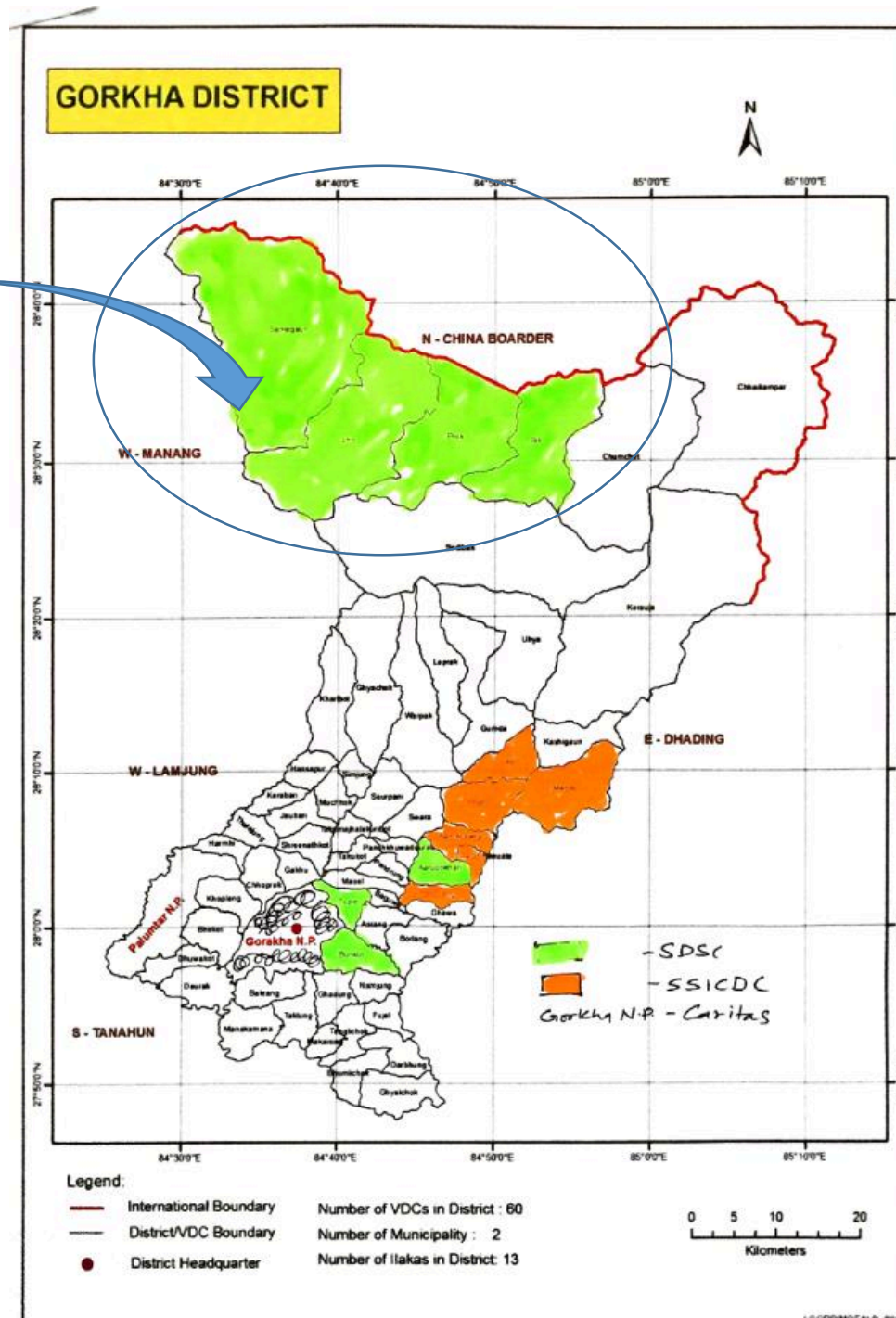
Northern VDCs

Samagaun

Lho

Prok

Bihi



Objective of Presentation

- To ensure houses are built back safer.
- The affected population has the knowledge and capacity to design and build appropriate , safe & acceptable housing solution.
- To involve indigenous materials & vernacular construction techniques.
- To identify some good practices and finalize recommendations to inform programming & designs suitable to the area.

Overview of Northern VDCs

- Population & Household (National Population Census 2011).

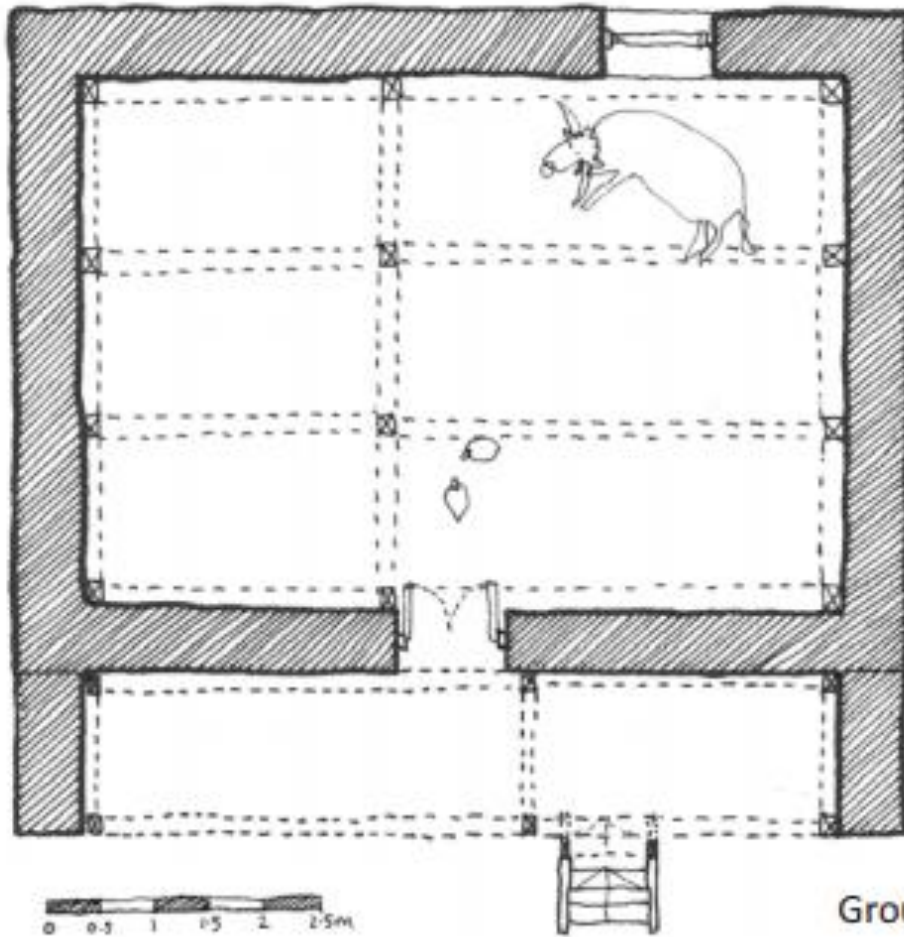
Gorkha Population : 272,169

VDC	Population	Household
Samagaun	604	197
Lho	711	256
Prok	575	187
Bihi	612	208
Total	2502	848

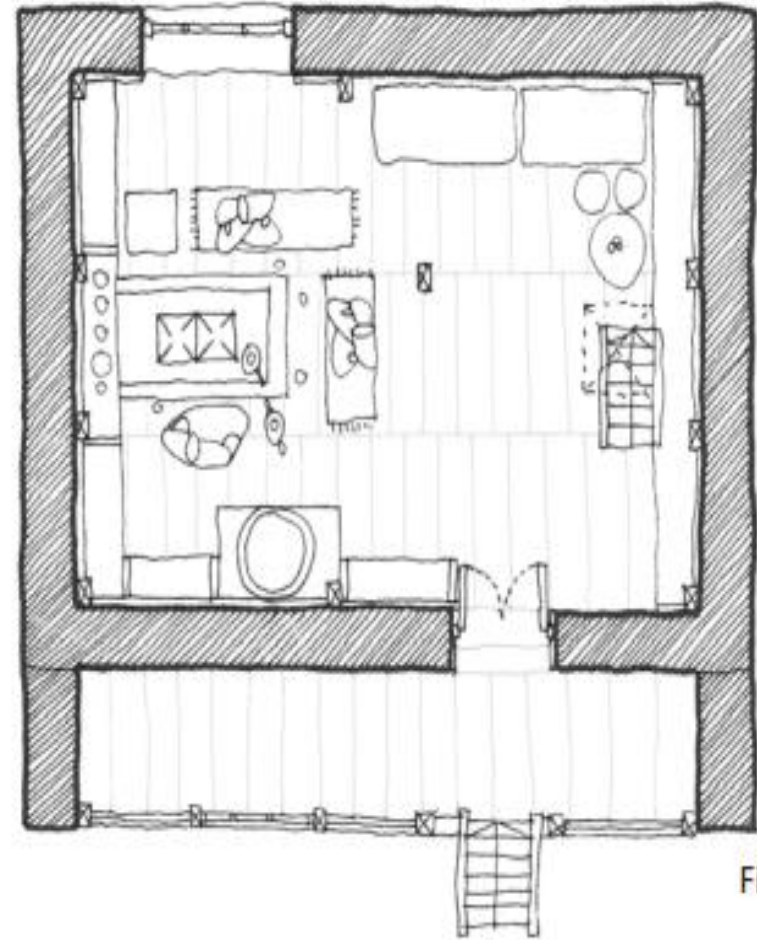
4 VDCs Population < 1% of total Gorkha
Current VDCs Statistics : approx. 4000

- Altitude ranges from 2000 m to more than 4000m above the sea level.
- Very cold weather through out the year and heavy snowfall in winter.
- There are no motorable roads. Mostly goods are transported by Humans and Mules using difficult foot trails or by very expensive air transport (helicopters).
- Livelihood depend on Agriculture, Tourism-Hotels, Trading & Carpentry.

Local House(Khim) Design : General Plan



Ground Floor Plan



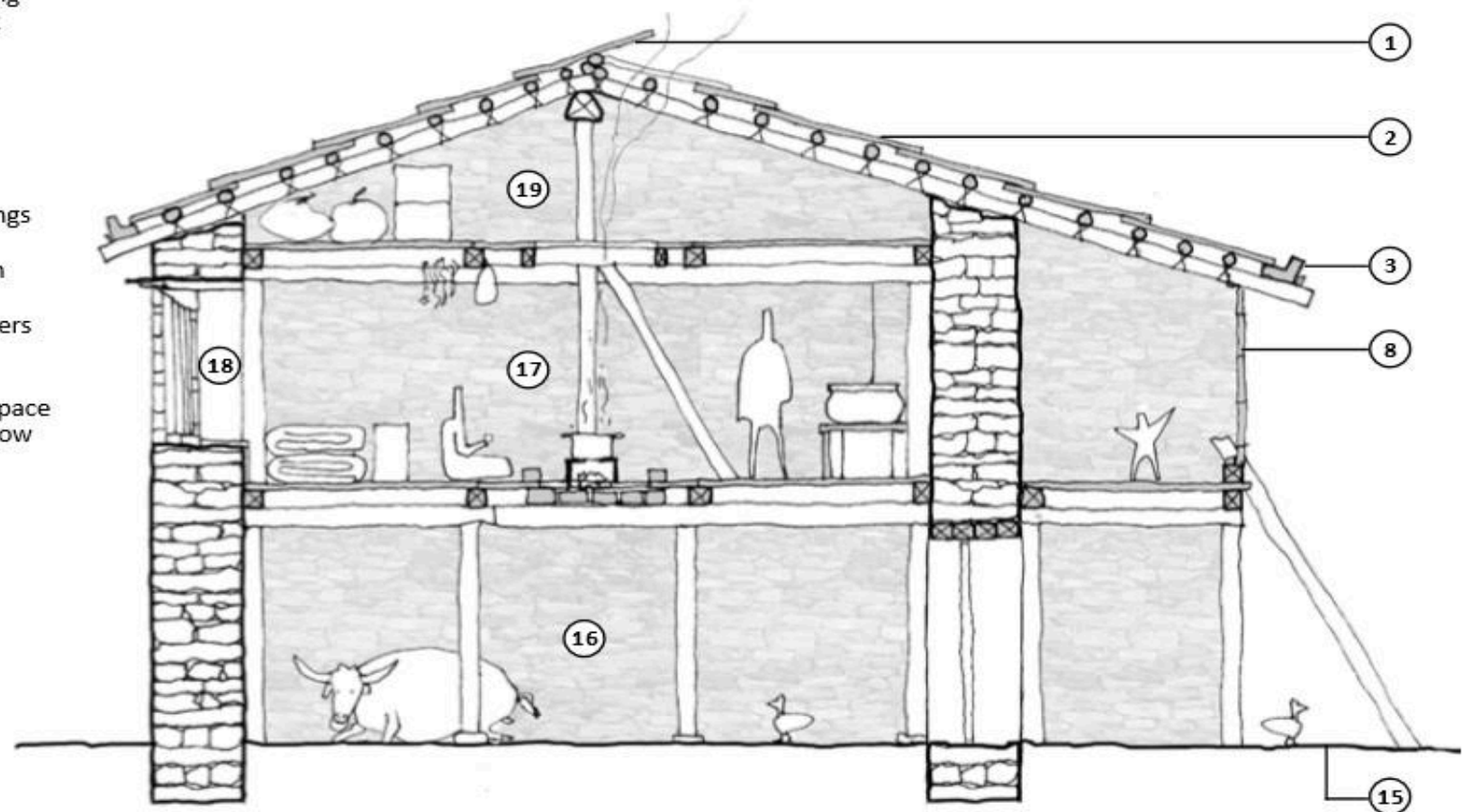
First Floor Plan

Local House(Khim) Design : General Elevation

3.2 Typical Khim (house) Study

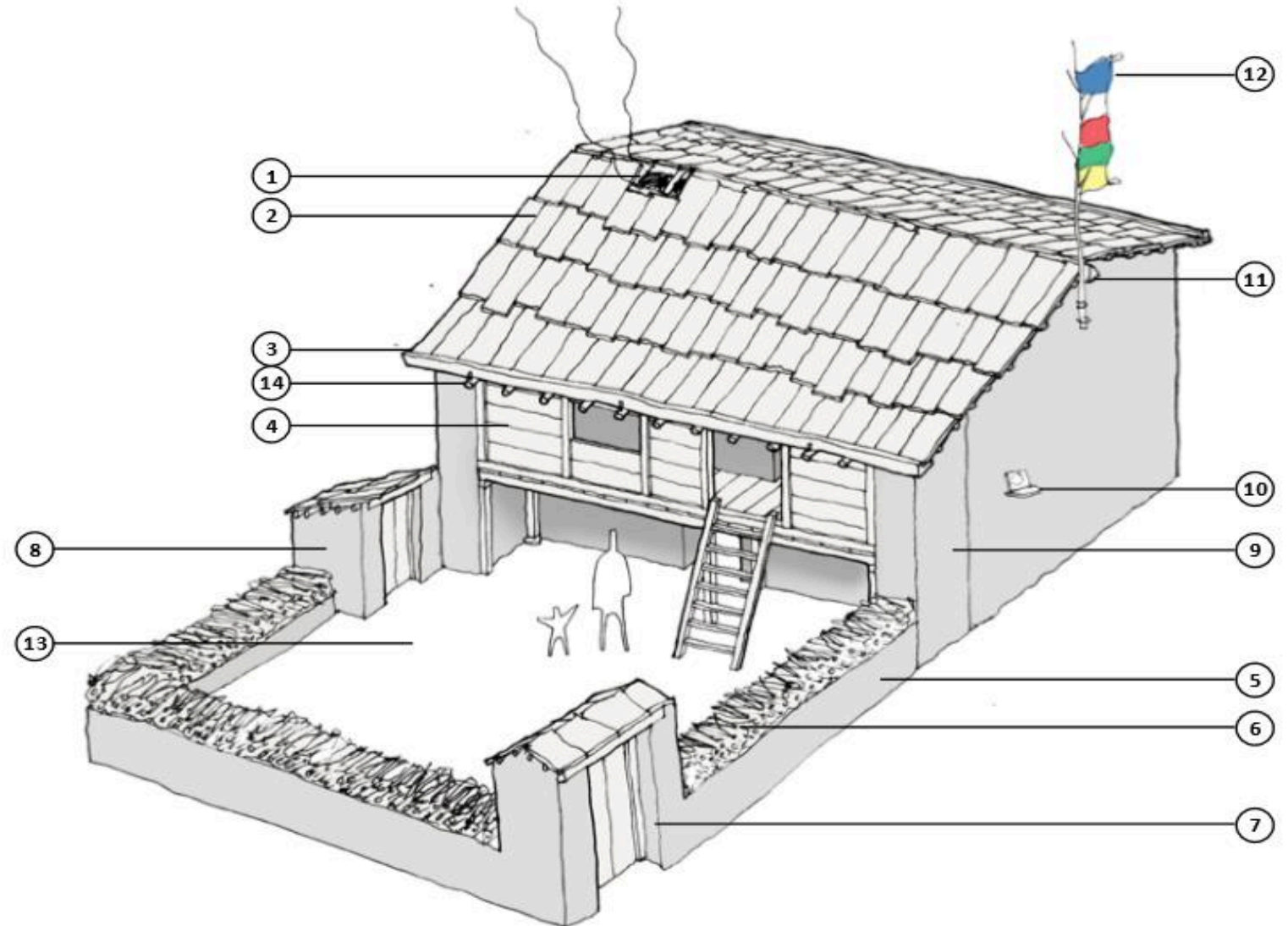
SHORT SECTION

- 1 - Roof void chimney
- 2 - Timber planks or stone tile roof covering
- 3 - Timber gutter carved from single trunk
- 4 - Timber verandah
- 5 - Boundary dry-stone wall
- 6 - Stored fire wood
- 7 - Entrance area
- 8 - Outside WC
- 9 - Separate stone wing wall to verandah
- 10 - Projecting stone shelf for tablet carvings
- 11 - Timber ridge beam
- 12 - Timber flag pole lashed to ridge beam
- 13 - Yard space for livestock
- 14 - Timber pegs fixed to support the gutters
- 15 - Compacted earth floor
- 16 - Ground floor livestock space
- 17 - First floor living / sleeping / cooking space
- 18 - Ornamental carved and painted window
- 19 - Attic for storage



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Settlement pattern in the North villages (Lho)



Typical dry stone House(Khim) : Photo



Major Damage (Pattern) to Local Houses by 2015 Earthquakes :

- Collapse of Gable end wall due to absence of through stones.
- Collapse of Verandah flank wall, these members are never keyed in to the main building wall.
- Top of wall collapsed around window and openings due to undersized lintels.
- Collapse of roof structure due to direct support on dry stone wall.
- Collapse of walls due to poor stone works , its shape and size.

Damages to houses by 2015 Earthquakes : In Photos



Callapsed gable end wall due to lack of through stones



Verandah flank wall collapse, these elements are never keyed in to the main building walls

Damages to houses by 2015 Earthquakes



Top of wall collapsed around a window opening, lintels are often undersized to take stone wall above



Collapsed roof structure

Main Challenges in the North

- Access(Only mode of transportation; mule, air transport and by foot).
- Construction material(Only Stone and Timber(soft wood) are available).
- Government subsidy is not enough (As per Locals)
- Timber not available in Samagaun VDC.
- No any Northern based guidelines in Building codes and government 17 Designs.
- Dry Stone masonry without any seismic bandings.
- Language barrier (IEC materials).
- Local acceptance and Tradition (relate to Building Design, Culture and Architecture).
- No any local treatment practices for timber.
- Seasonal Variation and Migration (Due to Snow, Wind and Rain).
- 70 % Houses are already constructed (Need Retrofitting solutions).

Suggestion & Conclusion

- Retrofitting Solutions to already constructed Houses (70% built).
- Focus on Northern Designs(Dry Stone masonry with Timber bandings)
- Design with Community consultation which reflects local architecture, Traditional Living spaces, Two story with Animal compartment in ground floor.
- IEC(information, Education and Communication) Materials all guidelines should be translated in local/friendly language.
- Availability and Supply chain of construction materials (And Rates).
- Top off in government subsidy for rural mountainous region.
- Engineers, technical support team and social support team should be able to interact with them (Language, tradition, behavior).



Stone banding, peice laid approx 1m above GL and is 100x deep x 2.5m long



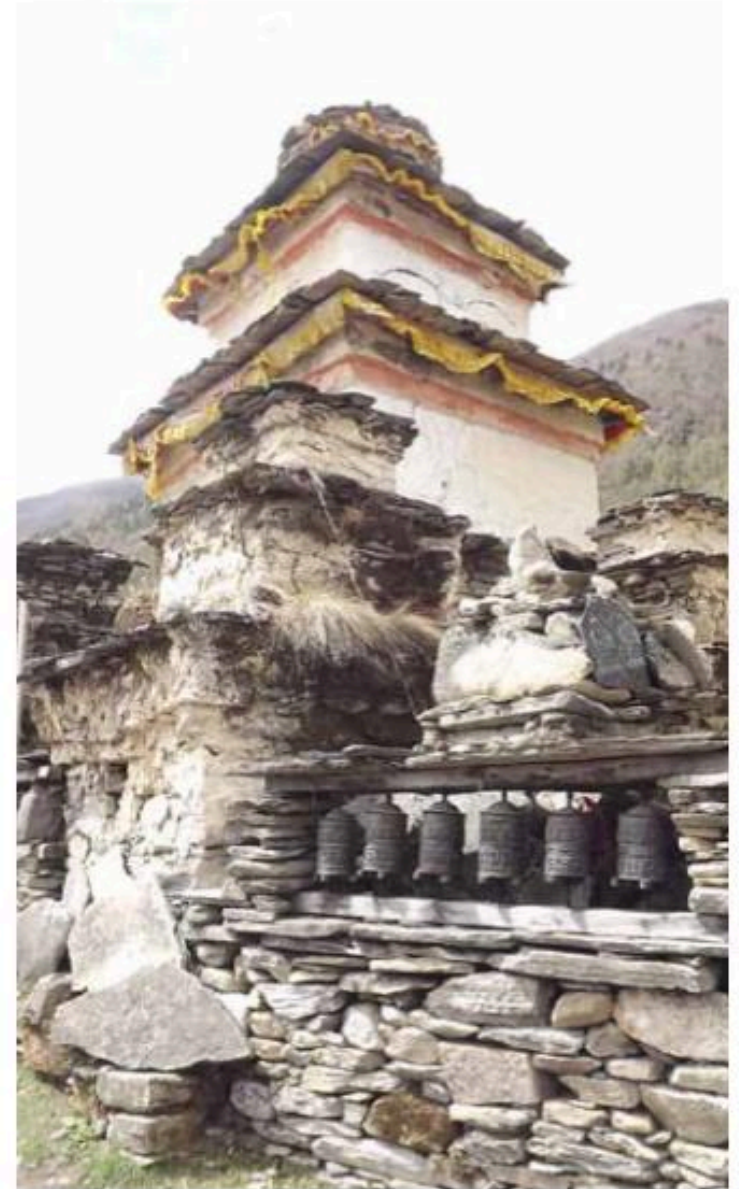
Use of steel pole and hooks on a concrete block hotel, wire wall ties used to secure the rafters



Carpenters working with hand tools



Best examples of stonework were observed on the hotel buildings



Dry-stone used for all the monument structures throughout the village



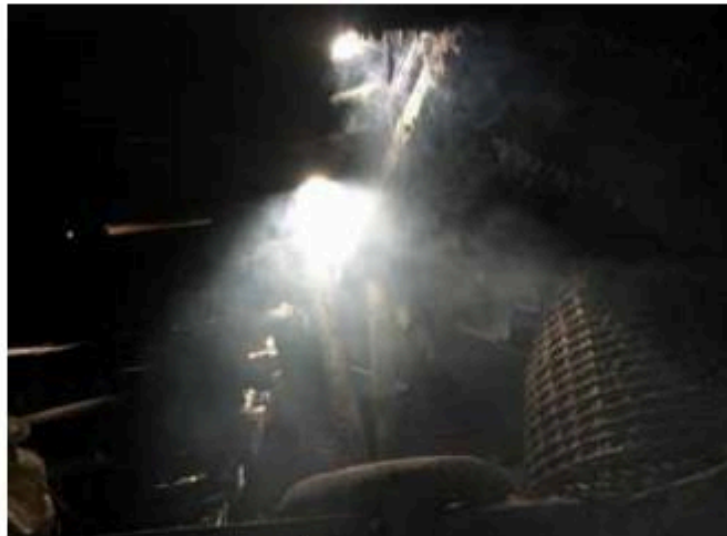
Timber post added to support the ridge beam



Timber roof planks as shingles



Large first floor extension in timber



Voids in the roof tiles/ shingles act as chimney's and provide an inadequate escape route for the smoke



Terraces of 3/4/5 Khims are found throughout the village with shared stone party wall



Mules transporting cement through the village to supply the hotel building projects



First storey timber found on an hotel structure



Good stonework again found on the hotel buildings, good coursing and stone shape and size



Timber single storey home



Hotel with new first floor timber floor

End of Presentation on Northern VDCs

Nepal Masonry and Earth Mortar Research

Oxford Brookes University
Catholic Relief Services
Purdue University

Earth Mortar in Masonry Construction



Earth mortar is an integral part of masonry construction in most of the affected areas. There is little information being provided about the quality of mortar in the guidelines or building code. CRS saw an opportunity to investigate this.

Earth Mortar in Masonry Construction



Issues with mortar are evident.

- Excessive proportion of mortar.
- Quality of soils for mortar.
- Issues of micro-cracking due to high clay content and quantity of water.
- Lack of tensile strength.

Earth Mortar in Masonry Construction - Improvement



- As part of the initial investigation, interns and volunteers with CRS made a control mix of mortar and added different fibers to the mix.
- After various simple trials , a laboratory test was conducted in Oxford Brookes University.



Some Highlights of Laboratory tests

Aim of Research :

to investigate the impact of various forms of reinforcement on the strength and integrity of mortar in traditional buildings (stone and mud mortar) and to present recommendations on how the mortar may be improved in practice.

Test samples mud mortar were provided with the following types of reinforcement:

None (control)

Rope

Straw

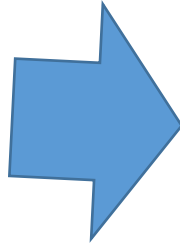
Shredded tarpaulin

Some Highlights of Laboratory tests

Two types of test were undertaken:

Test to initial failure (drop in load)

Test to total failure (sample breaks in two)



Conclusions:

- Significant variability in the initial resistance (when the samples first cracked) and ultimate failure was observed.
- But there was no clear distinction between the three types of reinforcement and two types of mortar.
- The performance of the samples between initial cracking and ultimate failure was largely dependent on the density of the reinforcement fibres and their location rather than the type of fibre or mortar.
- Samples with sparse fibres or those with fibres only on the top or bottom surface of the sample did not perform as well as those with many fibres located at mid-depth.
- Although the reinforcement did little to improve the shear resistance of the mortar, the improved post-cracking behaviour is likely to have some benefit in terms of the earthquake resilience of the masonry, by improving the structural integrity of the building and reducing the risk of total collapse during a seismic event.

Recommendation:

- Further research is needed to quantify the required reinforcement density and to provide guidance on achieving the required density and distribution in practice. Any further testing should be undertaken on cylinder samples and ultimately on small masonry panels subjected to dynamic loading.

Update :

Additional funding for shake table tests has been secured from existing relationships between CRS Head Office and Purdue University. Purdue will take the initial Brookes University results and test them further on a shake table, in order to compare the behaviour of the fibre reinforced mortar against non-fibre reinforced mortar in a small stone construction.

End of Presentation

Thank You !!